FACTORING A MAP OF COMPLEXES THROUGH A TERMWISE SPLIT INJECTION WITH A HOMOTOPY RETRACTION

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Editorial note. Counted span is the author's own proof environment, source lines 2865-2927, following the statement at 2839-2863.

Editorial note. No omissions: all four matrices displayed; the homotopy identity is verified in the adjacent remark the proof points to.

Allowed standard prerequisites (editorial).

- Homotopy computation remark. the remark immediately following this lemma in the same chapter (pointed to at ~2924: 'see computations in remark below'). Contains the matrix computation behind id - s pi = d h + h d.

Source reference. Original definition 014G: sources/stacks-derived/pinned-original-source.tex, source lines 2797–2797. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $\alpha : K^\bullet \to L^\bullet$ be a morphism of complexes of $\mathcal{A}$. There exists a factorization*

$$K^\bullet \xrightarrow{\tilde{\alpha}} \tilde{L}^\bullet \xrightarrow{\pi} L^\bullet, \qquad K^\bullet \xrightarrow[\alpha]{} L^\bullet$$

such that

(1) *$\tilde{\alpha}$ is a termwise split injection (see Definition 014G),*
(2) *there is a map of complexes $s : L^\bullet \to \tilde{L}^\bullet$ such that $\pi \circ s = id_{L^\bullet}$ and such that $s \circ \pi$ is homotopic to $id_{\tilde{L}^\bullet}$.*

Moreover, if both $K^\bullet$ and $L^\bullet$ are in $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$, then so is $\tilde{L}^\bullet$.

Proof. We set

$$\tilde{L}^n = L^n \oplus K^n \oplus K^{n+1}$$

and we define

$$d^n_{\tilde{L}} = \begin{pmatrix} d^n_L & 0 & 0 \\ 0 & d^n_K & \mathrm{id}_{K^{n+1}} \\ 0 & 0 & -d^{n+1}_K \end{pmatrix}$$

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In other words, $\tilde{L}^\bullet = L^\bullet \oplus C(1_{K^\bullet})$. Moreover, we set

$$\tilde{\alpha} = \begin{pmatrix} \alpha \\ \mathrm{id}_{K^n} \\ 0 \end{pmatrix}$$

which is clearly a split injection. It is also clear that it defines a morphism of complexes. We define

$$\pi = \begin{pmatrix} \mathrm{id}_{L^n} & 0 & 0 \end{pmatrix}$$

so that clearly $\pi \circ \tilde{\alpha} = \alpha$. We set

$$s = \begin{pmatrix} \mathrm{id}_{L^n} \\ 0 \\ 0 \end{pmatrix}$$

so that $\pi \circ s = \mathrm{id}_{L^\bullet}$. Finally, let $h^n : \tilde{L}^n \to \tilde{L}^{n-1}$ be the map which maps the summand K^n of $\tilde{L}^n$ via the identity morphism to the summand K^n of $\tilde{L}^{n-1}$. Then it is a trivial matter (see computations in remark below) to prove that

$$\mathrm{id}_{\tilde{L}^\bullet} - s \circ \pi = d \circ h + h \circ d$$

which finishes the proof of the lemma. $\square$